# Sample report: REF_GSM10113792

AstroGenetics: the physics of methylation, read on one sample

Specimen buffy coat · platform EPIC

**Research use only. Not a diagnostic test.** Each reading is printed with its commissioning status: *commissioned* means it passed a test sealed before its data was seen; a reading outside what an instrument was tested on is still shown, and says so. The full technical record (intake, calibration, tare, noise gate, safeguards, integrity, and how to run it yourself) is in the HTML report.

## At a glance

| Instrument | Question it answers | Reading | State | Status |
|---|---|---|---|---|
| **IAM-Deconvolver** | Which cells, in what shares | **CD4 T cells 27.3 %, Neutrophils 24.9 %, CD8 T cells 16.2 %** | | commissioned (neutrophils, monocytes, CD4 T, CD8 T) |
| **Met-A** | How far the neutrophil's tagging level has moved | **A 0.9902 (untared)** | untared: read A_rel (Stage T) | This score is preliminary until Met-A is officially commissioned on buffy coat. |
| **C-Score** | Are the departures clustered or scattered | **C 1.044 (untared)** | | This score is preliminary until the C-Score is officially commissioned on buffy coat. |
| **IAM-A** | Is each mark still held, against the sample's own reference | | | not run (no sequencing input) |

## AstroGenetics: the physics behind each reading

One law runs under every reading here: a mark held against thermal noise costs energy, and the loss of that hold shows as entropy, in bits. Each reading is a ratio to its healthy reference, so healthy reads 1. Its gauge runs from 0, the floor (below it a cell can no longer hold its identity), to saturation, the most the scale can register (every identity site fully scrambled). Both ends are derived from physics, not fitted to data.

| Instrument | What was measured | The physics |
|---|---|---|
| **Met-A** | ⟨H(β)⟩ over 5831 identity sites of the neutrophil, divided by its healthy expectation; **A = 0.990**. | **Gauge: 0 · healthy 1 · saturation 3.03.** Healthy 1 is the committed healthy reference of the cell type. In whole blood the expectation is the healthy profile of each cell type, weighted by this sample's cell shares. The floor (0 on the gauge) is the least entropy a cell can carry and still hold its identity; saturation is every identity site fully scrambled. |
| **C-Score** | How the departures of Met-A's identity sites cluster along the genome: the variance of the means of blocks of 10 neighboring sites over the variance of single sites, divided by the healthy median; **C = 1.044**. | A ratio of clustering, so it has no derived floor or ceiling. Healthy reads near 1; the Normal band 0.877 to 1.152 was set on 617 healthy arrays from 19 studies and held in two laboratories that had no part in setting it. The sky below shows the same departures. |

## Cell shares: IAM-Deconvolver against IAM-Atlas v2.1

| Cell | Share | Range (information) | Status |
|---|---|---|---|
| CD4 T cells | **27.3 %** | 24.9 to 29.0 % | commissioned |
| Neutrophils | **24.9 %** | 22.7 to 26.8 % | commissioned |
| CD8 T cells | **16.2 %** | 14.2 to 18.5 % | commissioned |
| Monocytes | **14.3 %** | 12.3 to 16.7 % | commissioned |
| NK cells | **10.0 %** | 8.9 to 11.4 % | This share is preliminary until the IAM-Deconvolver is officially commissioned on NK cells. |
| B cells | **7.3 %** | 6.5 to 8.2 % | This share is preliminary until the IAM-Deconvolver is officially commissioned on B cells. |

## C-Score on its bar

C-Score · buffy coat · not tared: no reading on the band

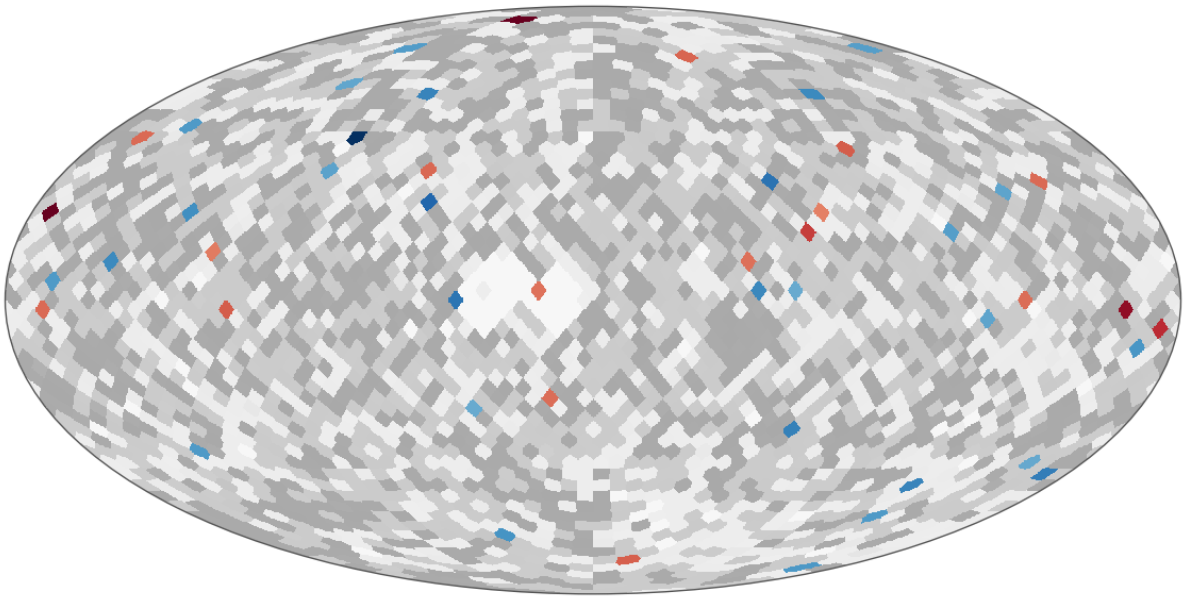



← scattered

clustered →

# What changed: the sample laid over its healthy reference

## What changed: this sample laid over healthy neutrophils



matches healthy (grey shade: the healthy cell's tagging)
more disorder than healthy: pattern lost (beyond +2 of its own spread)
more order than healthy (beyond −2)

Met-A 0.990  C-score 1.04  ·  48 of 3066 pixels changed

Grey is where this sample matches healthy neutrophils from IAM-Atlas v2.1 (the shade is the healthy cell's own tagging). Color marks only the sites that differ from healthy by more than 2 of this sample's own spread: red where its pattern is more disordered than healthy (marks lost), blue where it is more ordered. Every site keeps one place on a HEALPix sphere, set by its position in the genome, so a few scattered colored points are ordinary, and colored patches are changes that run along the genome, which is what the C-Score measures. A picture of the reading; nothing is scored from it.